

When an atom of sodium combines with an atom of chlorine, an electron is transferred from sodium atom to chlorine atom forming sodium chloride molecule thus both the atoms attain stable octet electronic configuration.

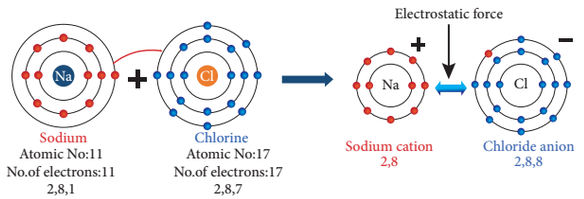


Figure. 13.6 Formation of ionic bond in sodium chloride

Illustration 2 – Formation of ionic bond in magnesium chloride ($MgCl_2$)

The atomic number of magnesium is 12 and the electronic configuration is 2, 8, 2. It has two electron excess to the nearest stable electronic configuration of a noble gas - Neon. So magnesium has a tendency to lose two electrons from its outermost shell and acquire a stable electronic configuration forming magnesium cation (Mg^{2+}).

As explained earlier two chlorine atoms will gain two electrons lost by the magnesium atom forming magnesium chloride molecule ($MgCl_2$) as shown in Figure 13.7.

Characteristics of Ionic compounds

The nature of bonding between the atoms of a molecule is the primary factor that determines the properties of compounds. By this way, in ionic compounds the atoms are held together by a strong electrostatic force that makes the compounds to have its characteristic features as follows:

Physical state: These compounds are formed because of the strong electrostatic force between cations and anions which are arranged in a well-defined geometrical pattern. Thus ionic compounds are crystalline solids at room temperature.

Electrical conductivity: Ionic compounds are crystalline solids and so their ions are tightly held together. The ions, therefore, cannot move freely, and they do not conduct electricity in solid state. However, in molten state their aqueous solutions conduct electricity.

Melting point: The strong electrostatic force between the cations and anions hold the ions tightly together, so very high energy is required to separate them. Hence ionic compounds have high melting and boiling points.

Solubility: Ionic compounds are soluble in polar solvents like water. They are insoluble in non-polar solvents like benzene (C_6H_6), carbon tetra chloride (CCl_4).

Density, hardness and brittleness: Ionic compounds have high density and they are quite hard because of the strong electrostatic force between the ions. But they are highly brittle.

Reactions: Ionic compounds undergo ionic reactions which are practically rapid and instantaneous.

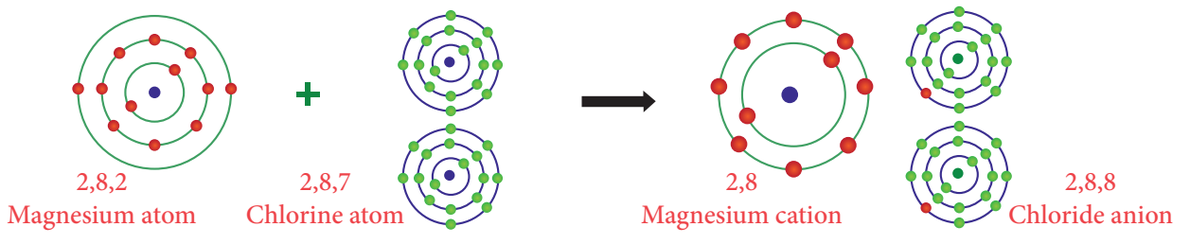


Figure. 13.7 Formation of ionic bond in magnesium chloride

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